Dheeraj Karki

MSc Robotics, University of Bristol, United Kingdom

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Professional Profile

MSc Robotics student at the University of Bristol with over ten years of software engineering experience and an academic background in electronics and embedded systems. Currently undertaking practical experiments involving Arduino-based mobile robots, infrared sensor characterisation, motion control, odometry and machine vision.

Experienced in CAD systems, computational geometry, experimental data analysis and technical problem-solving. Interested in applying engineering and programming skills to physical sensing, prototyping and experimental research.

Education

MSc Robotics **Sep 2026 – Sep 2027 (expected)**
University of Bristol *United Kingdom*

- Current study in robotic systems, fundamentals of robotics, machine vision and research methodology.

M.Tech in Software Systems **Jan 2017 – Jan 2019**
BITS Pilani *India*

- Specialisation: Embedded Systems. Work Integrated Learning Programme.
- CGPA: 8.6/10; First Division.
- Dissertation: CAD in Cloud.

B.Tech in Electronics and Communication **Aug 2008 – May 2012**
ICFAI University, Dehradun *India*

- CGPA: 9.42/10; Distinction.
- Thesis: Accelerated Parallel Processing Using OpenCL.

Research Interest

Multi-agent robotic systems; decentralised planning and decision-making; reliable information sharing under uncertainty; reinforcement learning; adaptive and trustworthy robotic autonomy.

Robotics and Academic Projects

Infrared Reflectance Sensor Characterisation *MSc Robotics | 2026*

- Designed and conducted experiments to investigate infrared reflectance sensor responses across different surfaces, strip widths and boundary conditions.
- Collected and analysed measurements to estimate effective sensor footprint and identify limitations in assumptions about sensor coverage.
- Investigated calibration, measurement repeatability and boundary effects to understand their influence on reliable robotic sensing.

Mobile Robot Raster-Scanning Project *MSc Robotics | 2026*

- Working with an Arduino-based mobile robot equipped with infrared reflectance sensors and wheel encoders.
- Investigating line-following control, rotation accuracy, wheel drift and positioning during raster-scanning experiments.
- Using sensor measurements, motor control and wheel odometry to evaluate robot behaviour and operational repeatability.

Machine Vision Experiments *MSc Robotics | 2026*

- Applied Python and OpenCV for image filtering, thresholding, segmentation, morphological operations and object detection.
- Worked with annotated image datasets and explored approaches to visual object and colour identification.

Beyond Binary Trust: Evidence-Aware Decision-Making *Independent Research | 2026–Present*

- Developed a synthetic multi-source information benchmark to investigate reliability and decision-making under uncertainty.
- Implemented and evaluated majority voting, global reliability weighting and domain-specific reliability weighting, using separate calibration and test datasets.
- Currently investigating evidence quality, provenance, source dependence and justified abstention in decision-making under uncertainty.

Technical Skills

Programming:	C++, C, Python, JavaScript
Sensors and Robotics:	Arduino-based robot experiments, infrared reflectance sensors, wheel encoders, odometry and motion control
Experimental Work:	Sensor characterisation, measurement analysis, calibration investigations, experimental repeatability and technical documentation
Machine Vision:	OpenCV, image filtering, thresholding, segmentation, morphological operations
CAD and Engineering:	ODA (Open Design Alliance), CAD systems, computational geometry, spatial data structures, interactive graphics and geometry processing
Software Engineering:	System architecture, performance optimisation, debugging, cross-platform development and code refactoring

Professional Experience

Arden Software **Jan 2019 – Sep 2026**
Software Development *India / Remote, UK*

Staff Software Developer Sep 2025 – Sep 2026

- Modernised a legacy desktop CAD application using a modern engine, supporting functionality across web, desktop and cloud platforms.
- Engineered quadtree-based spatial data structures for geometric queries and polygon-hierarchy resolution, improving performance by approximately 35%.
- Collaborated with engineering and design teams on complex technical problems, debugging and system improvements.

Senior Software Developer Oct 2021 – Sep 2025

- Designed reusable CAD annotation frameworks and implemented geometry-processing features for interactive CAD applications.
- Developed support for engineering drawing formats including DXF and DWG.

Software Developer Jan 2019 – Oct 2021

- Optimised HTML5 rendering pipelines through geometry caching and batched drawing operations, reducing rendering latency by approximately 90%.

Graebert GmbH **Feb 2016 – Jan 2019**
Software Engineering *Noida, India*

- Developed CAD editing tools, extensible plugin frameworks and interactive drawing systems.

Sevya Multimedia Technology **Feb 2014 – Jan 2016**
Software Developer *Hyderabad, India*

- Developed C/C++ automation tools for enterprise clients and an embedded home-automation prototype for customer demonstrations.

Electronics Corporation of India Limited **Mar 2013 – Jan 2014**
Engineer Consultant *Dehradun, India*

- Coordinated a team of ten operators on a National Population Register project.

Other Projects and Ventures

Frido – Productivity Application *Jul 2019 – Jul 2022*

- Co-owned and developed a productivity application for Apple's software ecosystem, collaborating with UI/UX designers.
- Released the application publicly and received recognition on Product Hunt.

Additional Projects and Achievements

- Co-owned and developed Frido, a publicly released productivity application for Apple's software ecosystem (2019–2022).
- Silver Medalist (2nd Rank), B.Tech, ICFAI University, 2012.
- Merit Fellowship Recipient (50% tuition waiver), ICFAI University, 2008–2012.